Mercury's precession due to Jupiter: a simple model

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***Abstract*—We estimate the precession of Mercury's orbit due to Jupiter's gravitational perturbation by numerically integrating the three-body problem (Sun–Mercury–Jupiter) using the Euler–Cromer method, extracting orbital elements via ellipse fitting at each perihelion passage. After compensating for the numerical precession of the integration scheme, we obtain $\dot{\omega} = 156.9554 \pm 0.0001''/$century, in reasonable agreement with the accepted value of $152.6''/$century given the simplifications of the model. We also characterise the synodic oscillations in eccentricity and semi-major axis and explain the observed offset between the mean perturbed eccentricity and the unperturbed value.**

I. Introduction

The study of planetary motion has been a cornerstone of astronomy and celestial mechanics since the times of Kepler and Newton. In particular, Mercury's orbit has been the subject of thorough analysis due to its peculiar precession, which cannot be fully explained by Newtonian mechanics. It has been shown that part of this precession is caused by gravitational interactions with other celestial bodies, with Jupiter being one of the main contributors due to its large mass. In this work, the problem of Mercury's precession is addressed by considering the influence of Jupiter within the framework of the three-body problem. To achieve this, numerical integration methods, such as the Euler-Cromer method, are implemented with the aim of reproducing the observed precession value and analyzing the accuracy of the proposed model.

II. Theory

A. The n body problem

We consider a system with n bodies interacting through Newtonian gravity. For the case where $n = 2$, and with a low enough energy ($\varepsilon < 1$), the theory predicts closed elliptic orbits, fixed with respect to the center of mass. The equation that describes them in polar coordinates is:[1]

$$r(\theta) = \frac{a(1-\varepsilon^2)}{1+\varepsilon\cos(\theta-\omega)} \tag{1}$$

where ε is the eccentricity of the orbit, a its semi-major axis, and ω the angle of the perihelion. These are known as the celestial elements of the orbit.[2]

However, for a value of $n \geq 3$ it is well known that a general analytic solution to this problem does not exist. Nevertheless, we can find solutions for cases where small perturbations are introduced in a two-body problem. According to perturbation theory, the solutions are elliptical orbits where the celestial elements slowly vary with time, i.e.:

$$r(\theta) \approx \frac{a(t)(1-\varepsilon(t)^2)}{1+\varepsilon(t)\cos(\theta-\omega(t))} \tag{2}$$

We are interested in the precession of the orbit, which is the variation of the perihelion angle with respect to time, $\dot{\omega}$.

This results in an overall precession of Mercury's orbit of $\dot{\omega} = 532''/$ century.

The planet with the highest contribution to this effect is Venus, due to its proximity, and the next one is Jupiter, due to its high mass.[3]

Our study will focus on the three-body problem given by Mercury, Jupiter and the Sun. Using Newtonian gravity, their interaction will be given by:

$$\vec{a}_M = -\frac{GM_S}{|\vec{r}_M-\vec{r}_S|^3}(\vec{r}_M-\vec{r}_S) - \frac{GM_J}{|\vec{r}_M-\vec{r}_J|^3}(\vec{r}_M-\vec{r}_J) \tag{3}$$

$$\vec{a}_J = -\frac{GM_S}{|\vec{r}_J-\vec{r}_S|^3}(\vec{r}_J-\vec{r}_S) \tag{4}$$

$$\vec{a}_S = -\frac{GM_J}{|\vec{r}_S-\vec{r}_J|^3}(\vec{r}_S-\vec{r}_J) \tag{5}$$

where the variables are the positions $\vec{r}_M, \vec{r}_J, \vec{r}_S$ of Mercury, Jupiter, and the Sun, and the parameters are described in table I.

We will consider the contributions of Mercury to the accelerations of the Sun and Jupiter negligible, since its mass is much lower than that of the other bodies.

It is important to note that we are using the astronomical system of units, based on the following quantities:

- Distance: Astronomical Unit (AU) $\equiv$ 149.597.870.700 m (distance from Earth to the Sun)
- Time: Earth year (year) $\equiv 365,25$ days
- Mass: Solar Mass ($M_\odot$) $\equiv 2\times10^{30}$ kg

B. Initial conditions

A useful approximation is that the orbits of each planet around the Sun can be viewed as two-body problems for short times. This allows us to obtain radii and eccentricities for the orbit of each planet around the Sun, and therefore obtain expressions for the initial conditions for each body.

In the case of the Solar System, all planets are located in the same plane, the ecliptic plane, which we will take as $z = 0$, and only use coordinates (x, y).

The initial conditions for Mercury will be its position in the perihelion, $\vec{r}_{min}$, and its speed at that point, $\vec{v}_{max}$, given by the following expressions:

$$r_{min} = a(1-\varepsilon) \tag{6}$$

$$v_{max} = \sqrt{GM_S\frac{1+\varepsilon}{r_{min}}} \tag{7}$$

where a is the semi-major axis of the orbit and ε is its eccentricity.

We will take as convention the initial position on the x axis and the initial speed perpendicular (as we are on the perihelion), pointing towards the y axis.

Jupiter's orbit will be considered circular. Its initial position will be also taken on the x axis, and its speed will be given by a uniform circular motion:

$$v_J = \frac{2\pi R}{T_J} \tag{8}$$

where T_J is Jupiter's period around the Sun. We will also consider the Sun's movement on this system, which will also occur at $z = 0$. We will take the Sun's initial position at the origin, but its initial velocity cannot be zero for convenience, as we want the center of mass to be static. The condition $\vec{0} = \vec{P}_{CM} = M_M \vec{v}_M + M_J \vec{v}_J + M_S \vec{v}_S$, implies that:

$$\vec{v}_S = -\frac{M_M \vec{v}_M + M_J \vec{v}_J}{M_S} \tag{9}$$

Finally, we will use Kepler's Third Law to obtain the period T of a planet given its semi-major axis a with the required precision:

$$T = 2\pi \sqrt{\frac{a^3}{G(M_S + M_P)}} \tag{10}$$

In this way, no numerical error is accumulated due to insufficient decimals.

C. *Mercury's precession*

According to observational data, Mercury's orbit precesses a very small amount, approximately $574.10 \pm 0.65''$/century.[4] Of this total, the contributions are:

- Celestial bodies: $\sim$ $532''$/century,[4] of which $\sim$ $152''$/century are due to Jupiter.[3]
- General Relativity: $43''$/century due to the multi-polar expansion of the resulting gravity potential.[4]

The aim of this paper is to recreate the value of Mercury's precession due to Jupiter using numerical simulations.

III. Numerical methods

We obtain the evolution of Mercury's orbital parameters due to Jupiter's perturbation with the following algorithm:

1) Integrate the equations of motion (3), (4), (5) using the Euler-Cromer method[5] as we need a method that conserves energy in each period to avoid accumulation of numerical error.
2) Calculate the polar coordinates of Mercury relative to the Sun: $r = \sqrt{(x_M - x_S)^2 + (y_M - y_S)^2}$, $\theta = \arctan((y_M - y_S)/(x_M - x_S))$.
3) Find the perihelia as relative minima of r.
4) Fit the data $(r(t), \theta(t))$ between the perihelia to an ellipse rotated by an angle ω, given by equation (1). From this, obtain the value of ω, a and ε for each period.
5) Add $2\pi n$ when necessary to compensate for the loops that the ellipse can make when rotating it.

This algorithm is run twice:

- First, with $M_J = 0$. Knowing that the expected orbital parameters the two body problem are $\omega = 0$, $a = a_0$ and $\varepsilon = \varepsilon_0$, we can calculate the deviation introduced by the method on each parameter.
- Then, with the real value of M_J. We subtract the deviations obtained in the previous step to calculate the real values of the orbital parameters.

IV. Results

The algorithm described in section III is applied to the three-body system over 500 orbital periods of Mercury (approximately 120 years), using $n_{\text{div}} = 5000$ time-step divisions per period (see table I).

A. *Precession of Mercury's perihelion*

Figure 1 shows the perihelion angle ω as a function of time. The behaviour is well described by a secular (linear) precession superimposed with synodic oscillations:

$$\omega(t) = mt + n - A \sin\left(\frac{2\pi}{T}(t - t_0)\right) + B \sin\left(\frac{\pi}{T}(t - t_0) + \phi\right) \tag{11}$$

whose best-fit parameters are listed in table II. The synodic oscillation period obtained from the fit is $T = 5.930$ years, consistent with half of Jupiter's orbital period, as expected from the geometry of their gravitational interaction:

$$\frac{T_J}{T_{\text{synodic}}} \approx 2, \quad \frac{T_M}{T_{\text{synodic}}} \approx 25$$

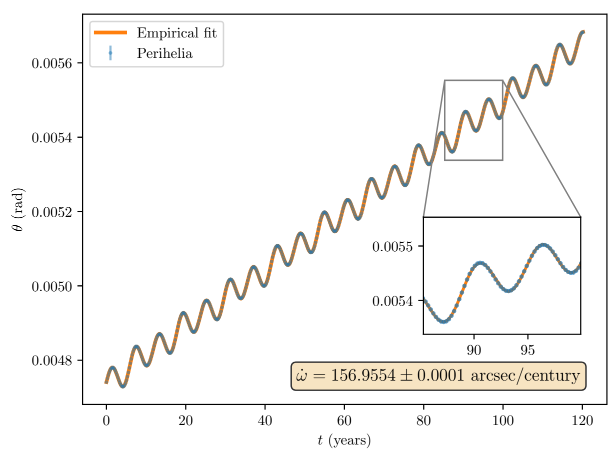

Fig. 1. Perihelion angle ω of Mercury as a function of time in the three-body problem. The inset shows the quality of the empirical fit at late times. The secular precession rate, after subtracting the numerical error from the integration scheme, is $\dot{\omega} = 156.9554 \pm 0.0001\,''$/century.

The secular precession rate, obtained from the slope m of the fit converted to arc seconds per century, is:

$$\dot{\omega} = 156.9554 \pm 0.0001\,''/\text{century} \tag{12}$$

The accepted value for Mercury's precession due to Jupiter is $152.6\,''$/century.[3] The $\sim 3\%$ discrepancy is attributable to the simplifications of the model: the assumption of a circular orbit for Jupiter, the restriction to a planar (2D) geometry, and minor higher-order effects accumulated over the 500-period simulation.

B. Evolution of eccentricity and semi-major axis

In addition to precession, Jupiter's perturbation induces periodic oscillations in Mercury's instantaneous eccentricity and semi-major axis. These are obtained from the ellipse fits at each perihelion passage after subtracting the systematic bias determined from the two-body calibration run.

Both quantities are well described by the empirical oscillatory model:

$$f(t) = A\sin\left(\frac{2\pi}{T}(t-t_0)\right) + B\sin\left(\frac{\pi}{T}(t-t_0)+\phi\right) + n \tag{13}$$

whose best-fit parameters are listed in tables III and IV. In both cases, the dominant oscillation period matches the synodic period of ~ 5.93 years found in the precession analysis.

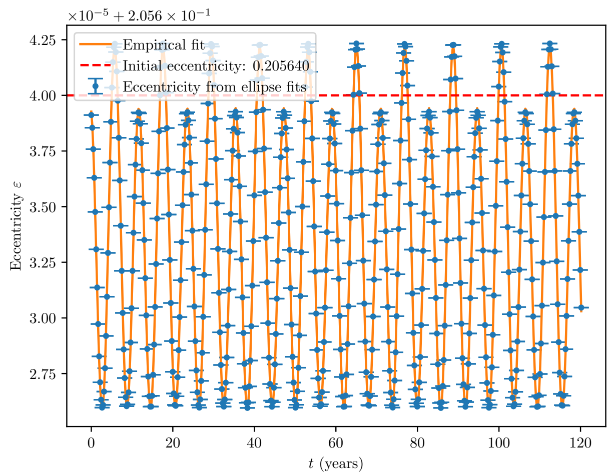

Fig. 2. Evolution of Mercury's orbital eccentricity due to Jupiter's perturbation. The orange curve is an empirical fit to equation (13). The dashed red line marks the unperturbed eccentricity $\varepsilon_0 = 0.20564$. The data oscillate around a mean value $\bar{\varepsilon} = 0.20563339$, systematically lower than ε_0 (see section IV-B1).

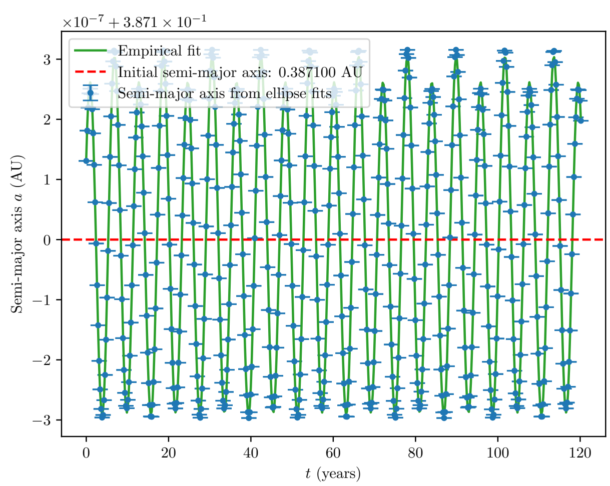

Fig. 3. Evolution of Mercury's orbital semi-major axis due to Jupiter's perturbation. The green curve is an empirical fit to equation (13). The dashed red line marks the unperturbed semi-major axis $a_0 = 0.3871$ AU. Unlike the eccentricity, the semi-major axis oscillates symmetrically around its initial value.

1) Eccentricity offset from the unperturbed value: A notable feature of figure 2 is that the mean eccentricity ($\bar{\varepsilon} = 0.20563339$) is systematically lower than the initial unperturbed value ($\varepsilon_0 = 0.20564$), by $\delta\varepsilon \approx 6.6 \times 10^{-6}$, which is comparable to the oscillation amplitude $A \approx 7.4 \times 10^{-6}$.

This offset is a physical effect, not a numerical artefact. The initial conditions place Mercury at the perihelion of its unperturbed Keplerian orbit, while Jupiter sits on the same radial axis at distance R. From the very first time step, Jupiter exerts a radial gravitational pull on Mercury that modifies the effective radial force balance. As a result, the eccentricity of the perturbed orbit is shifted from the bare value ε_0. In the language of perturbation theory, the time-averaged eccentricity is shifted by the perturbation, and the orbit oscillates around this *dressed* value rather than the original one.

The semi-major axis, by contrast, depends primarily on the total orbital energy, which is only modified to second order by the perturbation. This explains why the semi-major axis in figure 3 oscillates symmetrically around a_0, while the eccentricity does not.

V. Conclusion

Through numerical simulation of the Sun–Mercury–Jupiter three-body problem using the Euler–Cromer integrator, we have estimated the precession of Mercury's orbit due to Jupiter's gravitational perturbation.

After subtracting the numerical precession introduced by the integration scheme, calibrated via a two-body control run, the secular precession rate is:

$$\dot{\omega} = 156.9554 \pm 0.0001\,''/\text{century}$$

This is in reasonable agreement with the accepted value of $152.6''$/century, with the $\sim 3\%$ discrepancy attributable to the model's simplifications (circular Jupiter orbit, planar geometry).

Beyond precession, the simulation reproduces the synodic oscillations of Mercury's orbital elements due to Jupiter's perturbation, with a period of $T \approx 5.93$ years consistent with half of Jupiter's orbital period. The eccentricity and semi-major axis evolution are well described by a two-sinusoid empirical model. We have also provided a physical explanation for the observed offset between the mean perturbed eccentricity and the initial unperturbed value, attributing it to the gravitational dressing effect of Jupiter's perturbation.

We conclude that the Euler–Cromer method, combined with ellipse fitting and numerical error compensation, provides a computationally efficient and physically accurate approach for studying celestial mechanics problems of this type.

Appendix

For the simulation, we used the data enumerated in table I, taken from.[5]

Note that:

- The orbital period is recalculated using equation (10) to achieve more numerical precision.
- We integrate over 500 periods of Mercury because both Mercury's and Jupiter's orbits repeat every ~ 50 periods, so that the results obtained have a periodic pattern.

TABLE I
TABLE OF PARAMETERS USED IN THE SIMULATION.

Parameter	Symbol	Value
General parameters		
Gravit. const. ($\mathrm{AU}^3/\mathrm{year}^2/M_\odot$)	G	$4\pi^2$
Mercury's parameters		
Eccentricity	ε	0.20564
Semi-major axis (AU)	a	0.3871
Mass ($M_\odot$)	M_M	1.659×10^{-7}
Jupiter's parameters		
Radius of the orbit (AU)	R	5.2025
Mass ($M_\odot$)	M_J	9.542×10^{-4}
Simulation parameters		
Total time	t_f	$500T_M$

TABLE II
BEST-FIT PARAMETERS FOR THE PRECESSION MODEL (EQUATION (11)).

Parameter	Value $\pm$ Error
m (rad/s)	$(7.609413 \pm 0.000005) \times 10^{-6}$
T (s)	5.930144 ± 0.000001
A (rad)	$(-3.65252 \pm 0.00002) \times 10^{-5}$
B (rad)	$(-7.1828 \pm 0.0002) \times 10^{-6}$
t_0 (s)	-0.07834 ± 0.00001
ϕ (rad)	0.01159 ± 0.00003
n (rad)	$(4.7384108 \pm 0.0000003) \times 10^{-3}$

TABLE III
BEST-FIT PARAMETERS FOR THE ECCENTRICITY MODEL (EQUATION (13)).

Parameter	Value $\pm$ Error
T (s)	5.930195 ± 0.000001
A	$(-7.40057 \pm 0.00005) \times 10^{-6}$
B	$(1.36491 \pm 0.00005) \times 10^{-6}$
t_0 (s)	1.28335 ± 0.00001
ϕ (rad)	-0.80282 ± 0.00004
n	$0.20563339355 \pm 0.00000000003$

TABLE IV
BEST-FIT PARAMETERS FOR THE SEMI-MAJOR AXIS MODEL (EQUATION (13)).

Parameter	Value $\pm$ Error
T (s)	5.929990 ± 0.000003
A (AU)	$(2.86148 \pm 0.00006) \times 10^{-7}$
B (AU)	$(-2.1390 \pm 0.0006) \times 10^{-8}$
t_0 (s)	-0.55080 ± 0.00004
ϕ (rad)	0.7047 ± 0.0003
n (AU)	$0.387099996891 \pm 0.000000000004$

REFERENCES

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